AthleteLink AI

Complete Feature Architecture, Technical Walkthrough & Presentation Manual

VERSION 2.4
PRODUCTION

Confidential & Presentation-Ready

1. Executive Summary & Platform Architecture

AthleteLink AI is an intelligent sports talent discovery, scholarship management, and athlete development ecosystem. It bridges the critical divide between grassroots athletic prodigies across India and national sports federations, academies, coaches, and scouts. The technology stack utilizes a robust **Spring Boot 3.3.4 (Java 17)** backend with lightweight **Spring JDBC**, **MySQL 8.0** persistence, **JWT stateless authentication**, and a responsive frontend with interactive browser-based fitness testing engines.

Layer	Technology	Core Responsibility
Backend API	Spring Boot 3.3.4, Java 17, Spring Security	Stateless REST API, JWT auth filter, transaction control, input validation.
Data Persistence	MySQL 8.0, Spring JDBC Template	High-throughput schema with automated safe column/table self-healing.
Frontend Client	HTML5, Vanilla JS (ES6+), Tailwind CSS	Single-page dynamic application, high-fps reaction testing canvas, QR generator.
AI Talent Engine	Multi-variable dynamic algorithm	Computes dynamic talent index based on verified PRs, biometric consistency, and age.

2. Feature 1: Digital Athlete Scouting Card & QR Trial Gate Pass

A. Digital Scouting Card & Printable CV

FEATURE 1.1

Provides every registered athlete with an official, tamper-evident digital sporting CV. It showcases the athlete's verified bio-data, sport role, physical attributes, dynamic **AI Talent Index (/99)**, benchmark test history, and tournament honors.

- **Single-Page A4 PDF Printing Fixed:** Resolved the browser multi-page fragmentation issue. A dedicated `@media print` CSS stylesheet isolates the scouting sheet, switches dark elements to high-contrast paper formatting, hides non-printable modal buttons (`.no-print`), and ensures exact 1-page A4 printouts.
- **Phone QR Scanner & Wi-Fi Access:** Previously, localhost QR codes were unreachable on mobile phones. The system now automatically provides the laptop's Wi-Fi LAN IP (`10.65.24.238:8080`) and a 1-click toggle between **Phone on Wi-Fi**, **Laptop (localhost)**, and **Offline Digital vCard**. Any smartphone scanning the QR code instantly renders the verified card over the local Wi-Fi without internet or external hosting!

B. Official Trial Gate Pass & Reception QR Check-In

FEATURE 1.2

Eliminates paper tickets and unauthorized entry at athletic trial events. When an athlete books a trial slot, the platform generates a digital confirmation gate pass equipped with an encrypted Gate Pass QR code.

- **Venue Check-In Endpoint:** Scanning the QR code triggers `GET /api/trials/checkin?token=...`.
- **Database Mutation:** The booking record status changes from `CONFIRMED` to `CHECKED_IN` and logs the exact `checked_in_at` timestamp.
- **Automated In-App Notification:** Dispatches real-time attendance verification notifications to the athlete and academy trial director.

3. Feature 2: Athlete Performance Metrics & Personal Records (PR) Tracker

The **Performance Metrics & PR Tracker** allows athletes to record standardized athletic benchmarks across speed, endurance, power, and ballistic velocity.

Dynamic AI Talent Index Boost

In `PlatformService.java`, logging real athletic benchmarks dynamically influences the athlete's **AI Talent Index**. While incomplete profiles begin at a baseline of 40 points, verified performance logs contribute up to **+15 bonus points**:

```
int aiScore = 40; if (achievements.size() >= 1) aiScore += Math.min(35, achievements.size() * 10); if (metrics.size() >= 3) aiScore += 15; // Multi-discipline physical verification else if (metrics.size() >= 1) aiScore += 8; if (user.get("sport") != null) aiScore += 12; if (user.get("location") != null) aiScore += 8; if (trialCount > 0) aiScore += 5; aiScore = Math.min(99, aiScore);
```

Personal Records (PR) Intelligence: The system automatically detects whether a newly entered metric surpasses previous attempts. For sprint times, lower is better; for lifts and jumps, higher is better. Top records are surfaced as glowing badges in the dashboard.

Preset Benchmark	Discipline / Category	Standard Unit	Evaluation Logic
100m Sprint	Speed & Acceleration	Seconds (s)	Lower is better (e.g. 10.85s > 11.40s)
Vertical Jump	Explosive Power	Centimeters (cm)	Higher is better (e.g. 72cm)
Yo-Yo Recovery Test	Aerobic / Anaerobic Pacing	Level / Shuttles	Higher is better (e.g. 19.4)
Bench Press	Upper Body Strength	Kilograms (kg)	Higher is better (e.g. 110kg)
Bowling / Pitch Speed	Ballistic Rotational Velocity	km/h	Higher is better (e.g. 138 km/h)

4. Feature 3: National Athlete Skills & Fitness Drills Arena (Interactive)

Replaced static, theoretical problem statements with a **live, gamified physical skills testing arena** directly in the browser and field tracking interface.

1. 30-Second Rapid Reflex & Reaction Tap Test

A live interactive canvas drill. Random neural target stimuli appear across a specialized agility grid. Athletes tap targets as quickly as possible. The system tracks millisecond latency between stimulus and touch:

- **< 230 ms:** ⚡ Lightning Reflexes (National Elite Tier)
- **230 – 280 ms:** ⚡ Fast Reflexes (Pro Level)
- **> 280 ms:** ⚡ Quick Reflexes (Verified Baseline)

Athletes can 1-click submit their score to unlock verified badges and rank on the national leaderboard.

2. Core Stability Plank Endurance Stopwatch

An interactive, high-precision timer with milestone bells at 60s (Bronze), 120s (Silver), 180s (Gold), and 240s+ (Titanium Core). Integrates strict biomechanical form guidance cues.

3. 60-Second Jump Rope Cadence Tracker & National Leaderboard

Auditory and visual cadence pulse with a high-speed rep tap counter. Includes the **National Leaderboard** (`GET /api/challenges/leaderboard`) allowing scouts to sort and filter top-ranked athletes across all drills with verified video links.

5. Feature 4: AI Athletic Recovery & Injury Readiness Lab (New Section)

In modern high-performance sports, injury prevention and physiological workload management are the #1 determinants of athletic longevity. This brand-new dedicated section provides scientific daily recovery intelligence.

A. Daily Athletic Readiness Engine & AI Gauge

Athletes perform a 60-second morning check-in recording key physiological markers. The weighted algorithm computes the **Daily Readiness Index (0–100%)**:

```
// Weighted Algorithmic Model int sleepScore = (int) Math.min(100, (sleepHours / 8.0) * 100); int hrScore = (restingHr <= 52) ? 100 : (restingHr <= 60 ? 90 : (restingHr <= 70 ? 75 : 55)); int sorenessScore = (11 - sorenessLevel) * 10; int fatigueScore = (11 - fatigueLevel) * 10; int hydroScore = (int) Math.min(100, (hydrationLiters / 3.0) * 100); int readiness = Math.round(sleepScore*0.25 + hrScore*0.20 + sorenessScore*0.20 + fatigueScore*0.20 + hydroScore*0.15);
```

- **85 – 100% (Optimal):** Low Injury Risk. Cleared for maximal sprint speed and maximum power lifting.
- **65 – 84% (Moderate):** Moderate Risk. Prescribed technical and tactical drills; deload heavy eccentric lifts.
- **< 65% (Fatigue Warning):** Elevated Risk. Automated system notification triggered recommending active recovery, foam rolling, and contrast hydrotherapy.

B. Sport-Specific Injury Prehab & Parasympathetic Recovery Orb

- **Targeted Prehab Guidelines:** Select your discipline (Athletics, Cricket, Football, Badminton) to view evidence-based corrective exercises (e.g. Nordic Hamstring Lowers, Rotator Cuff Y-T-W, ACL Deceleration Landings) with clinical dosage recommendations.
- **3-Minute 4-7-8 Parasympathetic Recovery Breathing Orb:** An interactive visual breathing circle (Inhale 4s, Hold 7s, Exhale 8s) that triggers the vagus nerve to reduce post-workout cortisol and accelerate heart-rate recovery.

6. Presentation & Viva Demo Script (How to Explain to Others)

💡 **RECOMMENDED 3-MINUTE LIVE DEMO SEQUENCE**

1. **Step 1 – Dashboard & AI Talent Index:** Open `http://localhost:8080/#dashboard`. Point to the top banner showing *AI Talent Index* (e.g. 85/99). Explain that this isn't static text; it's dynamically evaluated based on physical test PRs and achievements.
2. **Step 2 – Digital Scouting Card & Wi-Fi QR:** Click *"Official Scouting Card & QR"*. Show the high-res printable CV. Click *"📱 Phone on Wi-Fi"* and show your phone camera scanning it over local Wi-Fi. Then click *"Print / Save PDF"* to demonstrate the single clean A4 page layout!
3. **Step 3 – Skills & Fitness Drills Arena:** Click *"Challenges & Arena"* in the nav bar. Click *"Start 30s Reflex Drill"*. Tap the targets live to demonstrate millisecond latency tracking. Hit *"Save to Profile"* to show instant badge unlocking and AI score boosting.
4. **Step 4 – AI Recovery Lab:** Click *"Recovery Lab"* in the nav. Adjust the resting HR and sleep sliders to show the live AI Readiness Gauge recalculating from 60% to 92%. Show the sport-specific prehab protocols and the 4-7-8 breathing timer.

7. Local Execution & Network Configuration

```
# 1. Run the pre-compiled JAR with your local MySQL password: cd "C:\Users\Shanmuga Priya\.gemini\antigravity\scratch\athletelink\backend" java -jar target/athletelink-backend-1.0.0.jar --
```

```
spring.datasource.password=YOUR_PASSWORD # 2. Access on Laptop: http://localhost:8080 # 3. Access on Mobile Phone (Connected to same Wi-Fi): http://10.65.24.238:8080
```